Adamya **Kumar**

Portfolio GitHub LinkedIn

Profile Summary

Distributed Backend Engineer with 5+ years of professional experience building scalable, high-availability backend services on AWS. Experienced in low-latency API development, event-driven processing using Kafka, Redis caching, and production monitoring with CloudWatch. Hands-on with microservices architecture, database modeling, CI/CD automation, and operating services in production environments.

Skill Set

- **Cloud (AWS):** EC2, API Gateway, Load Balancers, Target Groups, CloudWatch (logging & alerts), S3, Lambda (event triggers), VPC networking, IAM basics
- **Languages:** JavaScript (Node.js), TypeScript, SQL, Bash, C#
- **Backend & Distributed Systems:** Node.js, Express, Nest.js, REST APIs, Microservices Architecture, Event-Driven Systems, Kafka, GraphQL, Asynchronous Processing, Caching (Redis), Rate Limiting, API Design
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Data Modeling, Indexing, Query Optimization, Transactions
- **DevOps & Reliability:** Docker, Nginx, CI/CD Pipelines, Jenkins, GitHub Actions, Monitoring & Logging, Fault Tolerance, Production Debugging, Zero-Downtime Deployments
- **System Design:** Service-Oriented Architecture, Message Queues, Idempotency, Retry Mechanisms, Load Balancing, Horizontal Scaling
- **Additional:** Software Design Patterns, Agile Development, Shell Scripting

Recognitions

- Gold Medalist at Delhi Horse Show (2011, 2013)
- Recipient of Dada Saheb Phalke Film Award (2013)
- Research on Home Automation published in International Journal (Dec 2019)

Hobbies

Horse Riding, Game Development, Reading Tech Blogs, Music

Contact details

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Education

Masters in Business Administration (Finance) *from Gurukul Kangri, Haridwar.* **May 2019 – Apr 2021**

Bachelors of Technology (Computer Science) *from Gurukul Kangri, Haridwar.* **Jun 2015 – Apr 2019**

Experience

Hero MotoCorp | Backend Engineer *Gurugram* **Nov 2024 – Present**

- Designed and operated backend APIs behind API Gateway and load balancers to handle production traffic with fault-tolerant routing and zero-downtime deployments across environments.
- Introduced Kafka-based asynchronous workflows to decouple services and prevent cascading failures, improving system resilience during peak traffic conditions.
- Implemented Redis-backed session and caching layer to reduce repeated authentication and database load, lowering API latency by 20%.
- Established production observability using CloudWatch metrics, structured logging, and alerting to proactively detect failures and reduce incident resolution time.
- Built reusable internal Node.js libraries and standardized service patterns to improve maintainability and consistency across multiple backend services.
- Investigated and resolved event-consumer failures by adding validation and safe message handling, preventing processing outages and improving reliability of background jobs.

Appfoster | Software Engineer II *Noida, India* **Apr 2022 – Nov 2024**

- Designed and implemented end-to-end user workflows for a client engagement platform, building APIs and managing state transitions for messaging, scheduling, and automated follow-ups, improving user productivity by 35%.
- Integrated external services including LinkedIn and Microsoft subscription APIs with validation, retries, and asynchronous background workers to ensure reliable onboarding and billing automation, reducing manual effort by 40%.
- Built real-time applications using GraphQL and Next.js with backend event handling and task processing to support responsive user interactions and consistent data updates.
- Revamped a transaction-based marketplace backend by optimizing database queries, improving API response handling, and supporting concurrent user actions, increasing engagement by 20% and transactions by 15%.
- Developed a workflow-driven HR SaaS system managing approvals, role-based actions, and status tracking across multiple users, deployed behind reverse proxies and load-balanced instances, reducing HR process time by 25%.

Capline Services | Associate *Noida, India* **Aug 2021 – Apr 2022**

- Integrated 10+ external insurance provider APIs and built validation services to process and normalize incoming data, improving accuracy by 30%.
- Diagnosed recurring production issues using logs and monitoring metrics, identifying root causes and achieving 99.9% uptime while reducing support load by 15%.

ITDA | Training Desk Analyst *Dehradun* **Jan 2021 – May 2021**

- Supported rollout of digital E-Office systems by migrating 1000+ legacy records and coordinating with stakeholders for adoption.
- Automated manual file processing using Python scripts and batch workflows, reducing processing effort by 50% and paper usage by 40%.

Personal Learning Projects

Trading Algorithm: Nifty Options System

- Implemented an automated trading backend by processing real-time market data from Zerodha APIs and executing strategy decisions, enabling continuous analysis and automated trade execution without manual intervention.

Adaptive LMS (Learning Management System)

- Built a scalable SaaS learning platform by designing backend services with vector search and RAG pipelines to provide personalized recommendations and analytics-driven content delivery.